

-----flask-----

Install Flask->

- 1) `python --version`
- 2) `mkdir flask_project`
- 3) `cd flask_project`
- 4) `python -m venv venv`
- 5) `venv\Scripts\activate`
- 6) `pip install flask`
- 7) `pip show flask`

Your First Flask Application

`app.py`

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def home():
```

```
    return "Hello, Flask!"
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

```
python app.py
```

check your browser ----

What is a Route?

A route connects a URL to a Python function.

For example:

```
@app.route("/")  
def home():  
    return "Home Page"
```

```
@app.route("/about")  
def about():  
    return "About Page"
```

```
@app.route("/contact")  
def contact():  
    return "Contact Page"
```

Now:

http://127.0.0.1:5000/

↓
home()

http://127.0.0.1:5000/about

↓
about()

http://127.0.0.1:5000/contact

↓
contact()

This is one of the most important concepts in Flask.

GET-----

Example:

```
@app.route("/users", methods=["GET"])  
def get_users():  
    return "Get users"
```

POST:

```
@app.route("/users", methods=["POST"])  
def create_user():  
    return "Create user"
```

Flask REST API

Flask is commonly used to build REST APIs.

Example:

```
from flask import Flask, jsonify
```

```
app = Flask(__name__)
```

```
@app.route("/api/users")
```

```
def users():
```

```
    data = [  
        {  
            "id": 1,  
            "name": "Rahul"  
        },  
        {  
            "id": 2,  
            "name": "Aman"  
        }  
    ]
```

```
return jsonify(data)
```

```
if __name__ == "__main__":  
    app.run(debug=True)
```

Visit:

</api/users>

Response:

```
[  
  {  
    "id": 1,  
    "name": "Rahul"  
  },  
  {  
    "id": 2,  
    "name": "Aman"  
  }  
]
```

Dynamic URL

This is very important.

```
@app.route("/users/<int:id>")
```

```
def get_user(id):
```

```
    return f"User ID is {id}"
```

Complete Example->

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def home():
```

```
    return "Welcome to my Flask API"
```

```
@app.route("/about")
```

```
def about():
```

```
    return "This is the About page"
```

```
@app.route("/users/<int:id>")
```

```
def get_user(id):
```

```
    return f"User ID is {id}"
```

```
@app.route("/products")
```

```
def products():
```

```
    return "Product list"
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

Beginner Flask CRUD Project-----

```
from flask import Flask, request, jsonify
```

```
app = Flask(__name__)
```

```
students = []
```

```
@app.route("/")
```

```
def home():
```

```
    return "Student Management API"
```

```
# CREATE
```

```
@app.route("/students", methods=["POST"])
```

```
def create_student():
```

```
    data = request.get_json()
```

```
    student = {
```

```
        "id": len(students) + 1,
```

```
        "name": data["name"],
```

```
        "age": data["age"],
```

```
        "course": data["course"]
```

```
    }
```

```
students.append(student)
```

```
return jsonify({  
    "message": "Student created successfully",  
    "student": student  
}), 201
```

```
# READ ALL
```

```
@app.route("/students", methods=["GET"])
```

```
def get_students():
```

```
    return jsonify(students)
```

```
# READ ONE
```

```
@app.route("/students/<int:id>", methods=["GET"])
```

```
def get_student(id):
```

```
    for student in students:
```

```
        if student["id"] == id:
```

```
            return jsonify(student)
```

```
    return jsonify({
```

```
        "message": "Student not found"
```

```
}}, 404
```

```
# UPDATE
```

```
@app.route("/students/<int:id>", methods=["PUT"])
```

```
def update_student(id):
```

```
    for student in students:
```

```
        if student["id"] == id:
```

```
            data = request.get_json()
```

```
            student["name"] = data["name"]
```

```
            student["age"] = data["age"]
```

```
            student["course"] = data["course"]
```

```
            return jsonify({
```

```
                "message": "Student updated successfully",
```

```
                "student": student
```

```
            })
```

```
    return jsonify({
```

```
        "message": "Student not found"
```

```
    }}, 404
```

DELETE

@app.route("/students/<int:id>", methods=["DELETE"])

def delete_student(id):

for student in students:

if student["id"] == id:

students.remove(student)

return jsonify({

"message": "Student deleted successfully"

})

return jsonify({

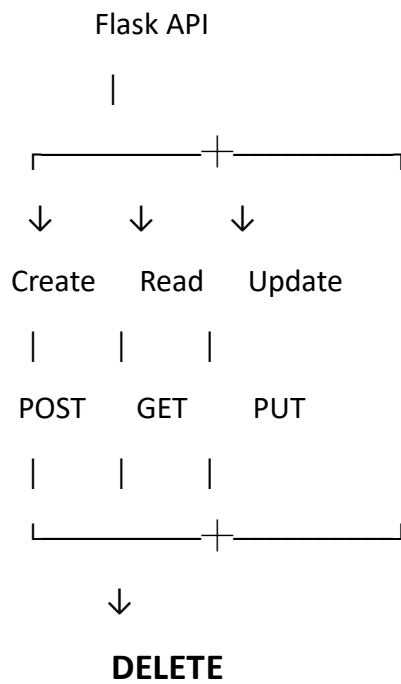
"message": "Student not found"

}), 404

if __name__ == "__main__":

app.run(debug=True)

How Students Should Understand It---



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